Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which statement best describes a reasonable physics approximation?

A. It is always acceptable to treat a human arm as a point particle when studying how the elbow joint rotates.
This confuses rotational motion (where size/shape matters) with translation of the center of mass.

B. When tracking a hiker's location on a map over several kilometers, modeling the hiker as a point mass is usually adequate.
Over kilometer-scale motion the object's size is negligible compared with the displacement, so a point-mass model is appropriate.

C. If you neglect air drag in a calculation, you are never allowed to describe the situation as "in vacuum."
This incorrectly treats "vacuum" as forbidden language rather than a modeling idealization used to ignore air effects.

D. None of the above is correct.
Option B is a standard and valid approximation for large-scale position changes.

Q2. Which of the following equations is correct?

A. 1 nickel = 1 $
This mistakes a 5-cent coin for a full dollar.

B. 1 nickel = 0.5 $
This incorrectly treats a nickel as fifty cents instead of five cents.

C. 1 nickel = 0.05 $
A dollar is 100 cents and a nickel is 5 cents, so $5 = 0.05\,\$$.

D. 1 nickel = 5 $
This reverses the conversion and produces a value that is orders of magnitude too large.

Q3. Which option lists an incorrect SI unit?

A. Momentum: kilogram·meter/second
Momentum has SI units $\mathrm{kg \cdot m/s}$, so this pairing is correct.

B. Power: joule/second
Power is energy per time, so J/s (watt) is correct.

C. Pressure: newton·meter
Pressure is force per area ($\mathrm{N/m^2}$), so $\mathrm{N \cdot m}$ is incorrect and corresponds to work/torque.

D. Acceleration: meter/second^2
Acceleration is the time-derivative of velocity, giving units $\mathrm{m/s^2}$.

Q4. A scout leaves the old watchtower located at the origin and spots an enemy camp at the coordinates $\vec{r} = -2\,\hat{\imath} + 7\,\hat{\jmath}$, where distance is measured in kilofurlongs. Which statement is correct?

A. The distance from the watchtower to the camp is 9 kilofurlongs.
This is the mistaken sum of the magnitudes of components $(2 + 7)$ rather than the vector magnitude.

B. The distance from the watchtower to the camp is 7 kilofurlongs.
This incorrectly uses only the y-component as the full distance.

C. The distance from the watchtower to the camp is 2 kilofurlongs.
This incorrectly uses only the x-component's magnitude as the full distance.

D. None of the above is correct.
The distance is $\sqrt{(-2)^2 + 7^2} = \sqrt{53} \approx 7.3$ kilofurlongs, which is not listed in A–C.

Q5. If the displacement of an object is given by

$$\vec{s} = \left((0.90\,\text{m/s}^3)t^3 - (1.25\,\text{m/s}^2)t^2\right)\hat{\imath},$$

which statement is correct?

A. This motion has constant acceleration.
Because $\vec{s}$ contains a t^3 term, the second derivative depends on t, so acceleration is not constant.

B. The magnitude of the acceleration is always 1.25 m/s^2.
This incorrectly treats the t^2 coefficient as the acceleration instead of differentiating twice.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s^2.
Differentiating twice gives a nonzero constant term, so $a(0) \neq 0$.

D. The magnitude of the acceleration at $t = 0$ is 2.50 m/s^2.
Since $a(t) = d^2s/dt^2 = 5.40t - 2.50$, at $t = 0$ the magnitude is $|-2.50| = 2.50$ m/s^2.

Q6. An object takes 4 s to move from position A, $\vec{r}_A = 7.0\,\hat{\imath} - 1.0\,\hat{\jmath}$, to position B, $\vec{r}_B = -1.0\,\hat{\imath} + 5.0\,\hat{\jmath}$, along a straight path (meters). Which statement is correct?

A. The distance traveled is 10.0 m.
The straight-line distance is $|\Delta\vec{r}| = \sqrt{(-8)^2 + (6)^2} = \sqrt{100} = 10.0$ m.

B. The speed at $t = 0$ must be 2.5 m/s.
Only total displacement and time are given, which fixes average speed but not the initial speed.

C. The acceleration at $t = 0$ must be 0.63 m/s^2.
Acceleration cannot be determined from just two positions and a travel time without additional motion details.

D. None of the above is correct.
Option A is directly determined from the displacement magnitude.

Q7. A basketball is launched with speed v_0 at an angle θ above the horizontal. What is the ball's speed at its maximum height (neglect air resistance)?

A. v_0.
This ignores that the vertical component of velocity decreases to zero at the top.

B. $v_0 \cos\theta$.
At maximum height the vertical velocity is zero, so the speed equals the unchanged horizontal component $v_0 \cos\theta$.

C. $v_0 \sin\theta$.
This mistakenly takes the initial vertical component as the speed at the top, even though it becomes zero there.

D. 0.
Only the vertical component becomes zero at the top; the horizontal component generally remains nonzero.

Q8. A drone initially moves with velocity $\vec{v}_0 = (18.0\ \text{m/s})\,\hat{\imath}$. Its motors then produce a constant acceleration $\vec{a} = (1.50\ \text{m/s}^2)\,\hat{\imath} + (3.20\ \text{m/s}^2)\,\hat{\jmath}$. Which statement is correct?

A. The x-component of the drone's velocity stays constant.
A nonzero a_x changes the x-component of velocity over time.

B. The y-component of the drone's velocity stays constant.
A nonzero a_y causes the y-component of velocity to change from its initial value.

C. Both the direction of motion and the speed change.
Because acceleration has components in both x and y, the velocity vector changes in both magnitude and direction.

D. None of the above is correct.
Option C correctly describes the effect of having nonzero acceleration in two perpendicular directions.

Q9. Two point charges exert an electrostatic force on each other. If the magnitude of each charge is multiplied by 4 and their separation is multiplied by 2, what happens to the magnitude of the force?

A. It stays the same.
This ignores that scaling charges and distance generally changes $F \propto q_1 q_2 / r^2$.

B. It decreases by a factor of 2.
This incorrectly applies the distance scaling linearly instead of through r^2 and misses the charge scaling.

C. It increases by a factor of 2.
This undercounts the effect of multiplying both charges (which multiplies $q_1 q_2$ by 16).

D. It increases by a factor of 4.
Coulomb's law gives a factor 16 from charges and a factor $1/4$ from r^2, so net change is $16 \times \frac{1}{4} = 4$.

E. It increases by a factor of 16.
This counts only the charge scaling and forgets that increasing separation reduces force by r^2.

Q10. A planet's escape speed is given by $v_{\text{esc}} = \sqrt{\dfrac{2GM}{R}}$. Which statement is correct?

A. Whether an object can escape depends on the object's mass because heavier objects need a larger escape speed.
Escape speed depends on the planet's M and R, not on the mass of the escaping object.

B. If a planet's mass increases while its radius stays the same, the escape speed decreases.
Since $v_{\text{esc}} \propto \sqrt{M}$, increasing M increases escape speed.

C. If a planet's radius increases while its mass stays the same, the escape speed increases.
Because $v_{\text{esc}} \propto 1/\sqrt{R}$, a larger radius lowers the escape speed.

D. A planet with larger mass and smaller radius has a higher escape speed than a planet with smaller mass and larger radius.
The formula shows v_{esc} grows with M and shrinks with R, so larger M and smaller R both raise it.

Q11. A constant force $\vec{F} = (2.50\,\text{N})\,\hat{\imath} - (6.00\,\text{N})\,\hat{\jmath}$ acts on a particle while it moves along the x-axis from $x = 1.20$ m to $x = 6.80$ m. What work (J) is done by the force?

A. -33.6 J
This incorrectly includes the y-component as if there were vertical displacement along the x-axis motion.

B. 14.0 J
The displacement is $\Delta\vec{r} = (5.60\,\text{m})\hat{\imath}$, so $W = \vec{F} \cdot \Delta\vec{r} = 2.50 \times 5.60 = 14.0$ J.

C. 33.6 J
This magnitude comes from multiplying $|F_y|$ by Δx, which is invalid because F_y is perpendicular to the displacement.

D. 8.40 J
This results from using the wrong displacement (e.g., subtracting endpoints incorrectly or using the wrong force component).

Q12. A 3.2 kg cart moving at 1.8 m/s hits a bumper and rebounds with speed 0.9 m/s in the opposite direction. The contact time is 0.15 s. What is the magnitude of the average force on the cart during the collision?

A. 19.2 N
This corresponds to dividing mv_i by time and ignores the reversal of direction (the full change in momentum).

B. 38.4 N
This typically comes from using $\Delta v = 1.8 - 0.9$ instead of $\Delta v = 1.8 - (-0.9)$.

C. 57.6 N
With $v_i = +1.8$ and $v_f = -0.9$, $|\Delta p| = m|v_f - v_i| = 3.2 \times 2.7 = 8.64$ N s, so $|F_{\text{avg}}| = 8.64/0.15 = 57.6$ N.

D. 76.8 N
This overestimates the force by using an incorrect time or by adding speeds without applying the correct momentum change.

Q13. A 65 kg skateboarder is initially at rest. A friend throws a 0.45 kg medicine ball horizontally at 12 m/s, and the skateboarder catches it (the ball and skateboarder move together). What is the skateboarder's speed immediately after the catch?

A. 0.083 m/s
For a perfectly inelastic catch, $v_f = \dfrac{m_b v_b}{m_s + m_b} = \dfrac{0.45 \times 12}{65.45} \approx 0.083$ m/s.

B. 0.50 m/s
This is too large and typically comes from forgetting to divide by the combined mass.

C. 12 m/s
This incorrectly assumes the skateboarder takes on the ball's initial speed despite the much larger combined mass.

D. 1.0 m/s
This overestimates the result by treating momentum as if it were conserved without accounting for the skateboarder's large mass in the denominator.
